Leeseok Kim

Curriculum Vitae

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Personal Webpage
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Academic Appointments

2026–Present **Postdoctoral Researcher**, *University of New Mexico*, Albuquerque, NM.
Based at Los Alamos National Laboratory (Theoretical Division, T-4).
Mentor: Luis Pedro García-Pintos.

Education

2021– 2026 **Ph.D., Computer Engineering**, *University of New Mexico*, Albuquerque, NM, *with Distinction*.
Dissertation: Quantum Control Protocols for Robust Quantum Computing. Advisor: Milad Marvian.

2017–2021 **B.A., Physics and Mathematics**, *Cornell University*, Ithaca, NY.

Research Interests

Quantum computing, Quantum simulation, Open quantum systems, Quantum control.

Publications

2026 **L. Kim**, "Rapid mixing of quantum spin chains at any finite temperature." arXiv:2610.01190.

2026 **L. Kim and L.P. García-Pintos**, "Randomized product formulas beyond optimal deterministic scaling." arXiv:2608.07720.

2026 **L. Kim, F. Riberi, K. Baca, and M. Marvian**, "Randomized Quantum Optimal Control." arXiv:2607.10946.

2026 **L. Kim and M. Marvian**, "Robust high-order quantum simulation using finite-width pulses." arXiv:2603.15502.

2026 **L. Kim and M. Marvian**, "High-order dynamical decoupling in the weak-coupling regime." *Physical Review Letters* 137, 120801.

2026 **K. R. Dizaji, L. Kim, M. Marvian, and C. Arenz**, "High-order Zeno sequences." *Physical Review Research* 8(2), 023051.

2026 **C. Yi, L. Kim, and M. Marvian**, "Faster Randomized Dynamical Decoupling." *Physical Review Letters* 136, 010601.
Conference presentations: QSim 2025 [Recorded Talk] and SQuInT 2025.
Media coverage: UNM Newsroom; Phys.org (May 2026).

2024 **L. Kim, S. Lloyd, and M. Marvian**, "Hamiltonian Quantum Generative Adversarial Networks." *Physical Review Research* 6(3), 033019.
Conference presentations: AQC 2023.

2023 **L. Kim and J. L. Palacios**, "On the Moments of the Number of Hires in the Assistant Hiring Algorithm." *Algorithms* 16(2), 70.

2022 **T. Hur*, L. Kim*, and D. K. Park**, "Quantum convolutional neural network for classical data classification." *Quantum Machine Intelligence* 4(1), 3.
*Equal contribution.

Selected Talks

Faster Randomized Dynamical Decoupling

2025 **SQuInT 2025**, *PAIS, UNM*, Albuquerque, NM.

2025 **QSim 2025 [Recorded Talk]**, *IBM*, New York, NY.

Hamiltonian Quantum Generative Adversarial Networks

2023 **AQC 2023**, *PAIS, UNM*, Albuquerque, NM.

2023 **QAISG's QML Seminar [Recorded Talk]**, *Centre for Quantum Technologies, NUS*, Singapore (Invited, virtual).

Patent Applications

2025 **Faster Randomized Dynamical Decoupling [Link]**, *C. Yi, L. Kim, and M. Marvian*, U.S. Patent Application No. 19/339,205, (Filed: Sep 2025).

Professional Service

Peer review Journals: PRX Quantum, Quantum, PRA.
Conferences: QIP 2027.

(Last updated: Oct 2026)